

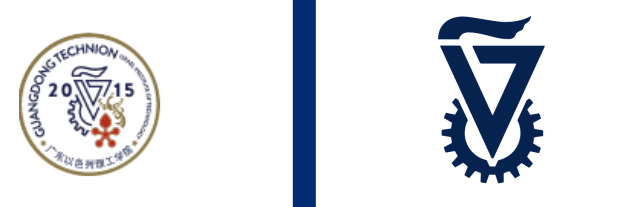
Propeller-Assisted Robust 3D Hopping Robot with Hierarchical Force Allocation

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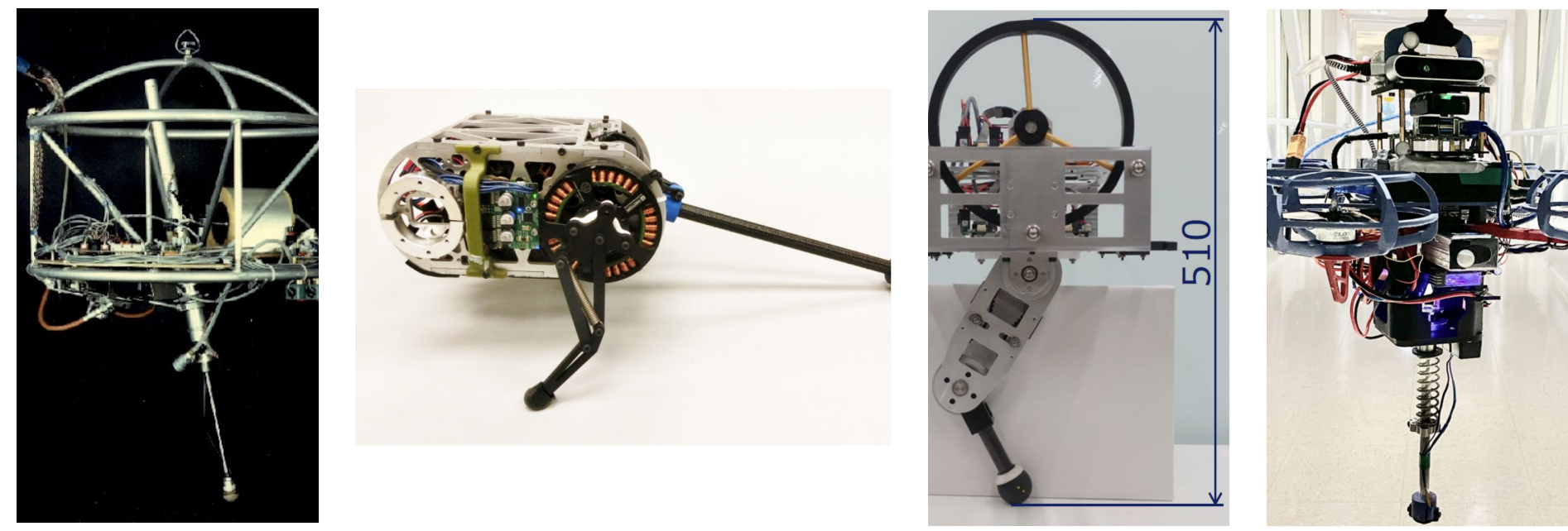
ACCEPTED PAPER



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1. Problem - hybrid dynamics create an authority gap



Stance

Ground reaction force exists only during a short contact interval.

Flight

No contact: attitude becomes underactuated.

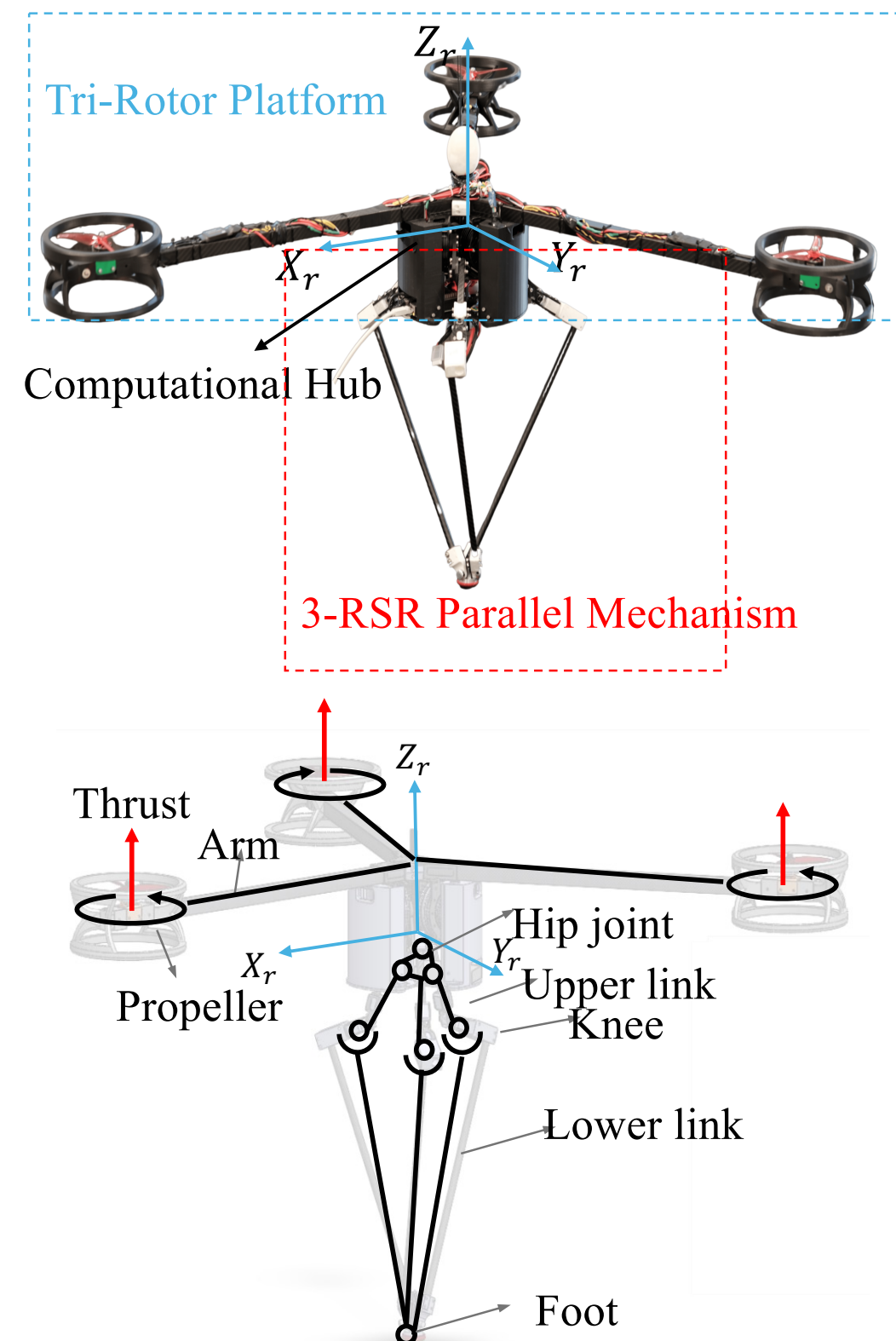
Disturbance

Impacts and pushes inject attitude and horizontal momentum.

Control authority disappears exactly when attitude can keep drifting.

Design question: How should leg contact force and rotor thrust be coordinated across stance and flight?**Target:** keep the leg primary and use rotors only where needed.

2. Platform - hardware built for leg-first hopping



3-RSR parallel leg

Three motor-driven RSR chains meet at one foot, enabling fast 3-DoF translational and omnidirectional placement with only **0.35 kg** moving leg mass.

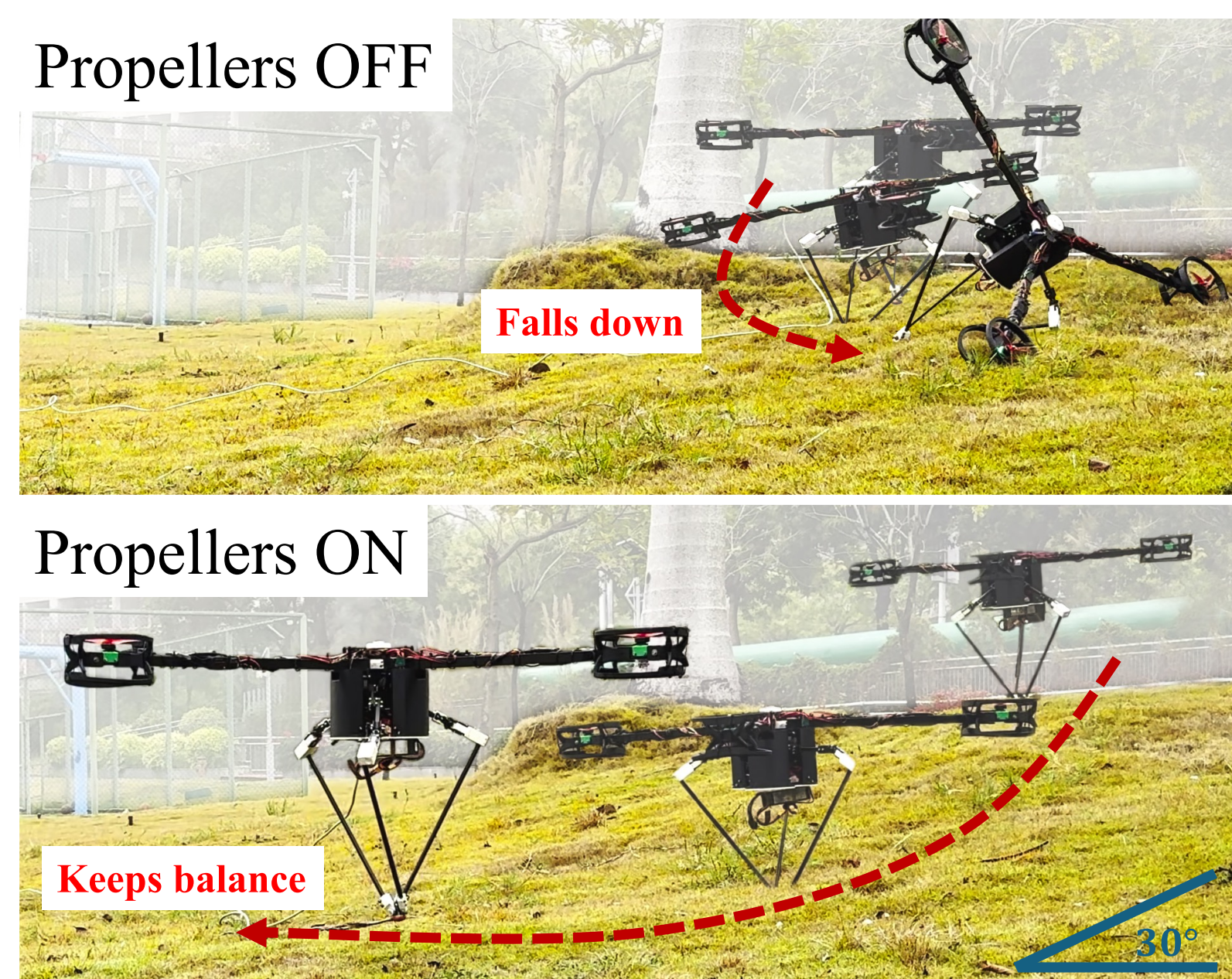
Computational hub

Batteries and electronics sit near the hip, lowering the center of mass and reducing rotational inertia.

Y-layout tri-rotor

Differential thrust provides continuous attitude authority. Each rotor reaches **10 N**, while total maximum thrust remains only **0.82 mg**: this is still a hopper, not a drone.**3.75 kg total** **9.2% leg mass** **20 Nm / leg motor** **500 Hz control****Light moving leg + rigid rotor assembly** preserves fast foot placement and supports the SRB approximation.

3. First proof - stable on 30° grass



Propellers off

After asymmetric contact, attitude error grows. With no flight-phase actuator, the robot falls.

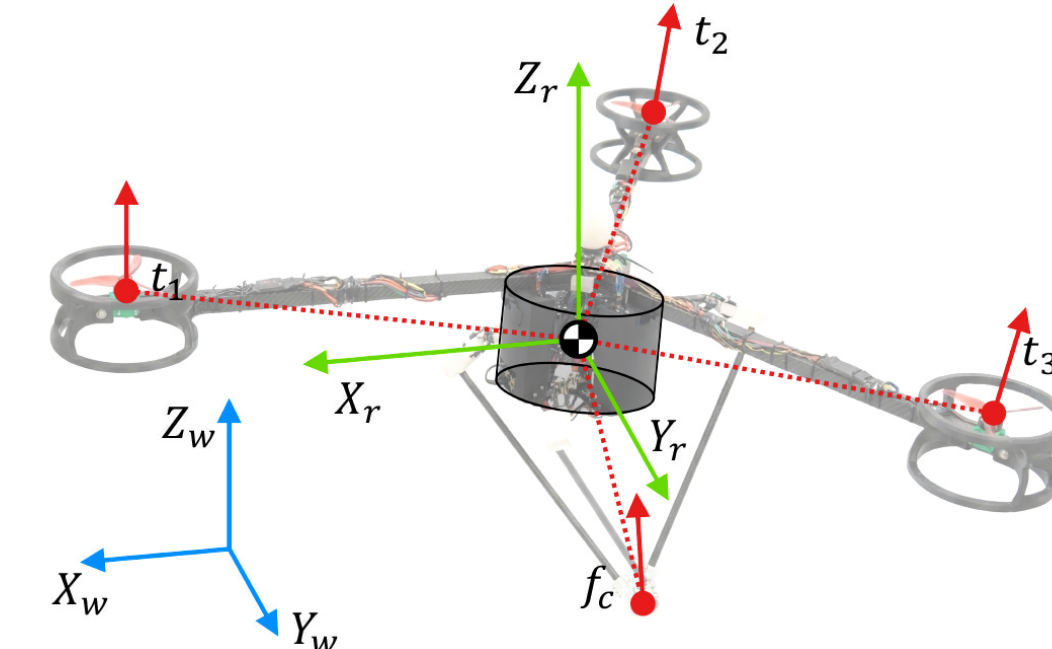
Propellers on

Differential thrust remains available through flight and restores attitude before the next landing.

What this establishes

Residual authority in stance.**Primary** attitude authority in flight.**Faster recovery** from uneven impacts.**Not primary lift:** the rotors close the attitude-authority gap.**Propellers off:** attitude diverges and the robot falls. **Propellers on:** it remains upright and keeps hopping.

4. Model - one rigid body, two actuator sets



Net wrench = leg contact + three rotors

$$\begin{bmatrix} \mathbf{F}_{net} \\ \boldsymbol{\tau}_{net} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_c \\ \mathbf{r}_c \times \mathbf{f}_c \end{bmatrix} + \sum_{i=1}^3 \begin{bmatrix} \mathbf{d}_i \\ \mathbf{r}_i \times \mathbf{d}_i \end{bmatrix} \mathbf{t}_i$$

Newton-Euler dynamics of the CoM

$$m\mathbf{a} = \mathbf{F}_{net} + m\mathbf{g}, \quad \dot{\boldsymbol{\omega}} = \mathbf{I}^{-1}(\mathbf{R}^T \boldsymbol{\tau}_{net} - \boldsymbol{\omega} \times \mathbf{I} \boldsymbol{\omega})$$

SRB actuation map: contact force $\mathbf{f}_c$ and thrusts $\mathbf{t}_{1,2,3}$ jointly produce the body wrench.

Symbols (values used in all experiments)

 $\mathbf{f}_c$: ground reaction force (world frame) $\mathbf{r}_c, \mathbf{r}_i$: arms: CoM $\rightarrow$ contact, rotor i $\mathbf{t}_i \in [0, 10]$ N thrust along $\mathbf{d}_i = \mathbf{R}\mathbf{e}_3$ $\mathbf{F}_{net}, \boldsymbol{\tau}_{net}$: net force, moment on CoM $m=3.75$ kg mass; $\mathbf{g}$ gravity; $\mathbf{I}$ inertia $\mathbf{R} \in \text{SO}(3)$ attitude; $\boldsymbol{\omega}$ body rate $\mathbf{J}$ leg Jacobian; $\tau_{max}=20$ Nm limit $\mu=0.4$ friction coefficient $p_z, \dot{p}_z$: CoM height, velocity $h_{des}=0.664$ m apex; $l_0=0.464$ m leg l leg length; Δ_{td} touchdown threshold k_z, b_z, k_F vertical, k_v, k_r Raibert, $\mathbf{K}_R, \mathbf{K}_\omega$ attitude gains

5. Control - leg-priority hierarchical force allocation

A. Vertical energy hopping (stance)

$$E_{sys} = \frac{1}{2} m \dot{p}_z^2 + m g p_z, \quad E_{des} = m g h_{des}$$

$$\mathbf{f}_{c,z}^{des} = \max(0, k_z(h_{des} - p_z) - b_z \dot{p}_z + k_E(E_{des} - E_{sys}))$$

Injects push-off energy, absorbs impact.

B. SO(3) attitude PD (both phases)

$$\mathbf{e}_R = \frac{1}{2} (\mathbf{R}_{des}^T \mathbf{R} - \mathbf{R}^T \mathbf{R}_{des})^\vee$$

$$\boldsymbol{\tau}_0^{des} = -\mathbf{K}_R \mathbf{e}_R + \mathbf{K}_\omega (\boldsymbol{\omega}^{des} - \boldsymbol{\omega})$$

Roll/pitch only; yaw left free. $\boldsymbol{\tau}^{des} = \mathbf{R} \boldsymbol{\tau}_0^{des}$.

C. Leg first - constrained least squares

$$\mathbf{f}_{c,xy}^{cmd} = \arg \min_{\mathbf{f}_{c,xy}} \|\mathbf{r}_c \times \mathbf{f}_c - (\boldsymbol{\tau}^{des})_{rp}\|_2^2$$

$$\text{s.t. } \|\mathbf{f}_{xy}\|_2 \leq \mu \mathbf{f}_{c,z}^{des}, \quad \|\mathbf{J}^T (-\mathbf{R}^T \mathbf{f}_c)\|_\infty \leq \tau_{max}$$

The leg realizes all it can under friction + torque limits.

D. Rotors take only the residual

$$\boldsymbol{\tau}_{res}^{cmd} = \boldsymbol{\tau}^{des} - \mathbf{r}_c \times \mathbf{f}_c^{cmd}$$

$$\delta \mathbf{t}^{cmd} = \arg \min_{\delta \mathbf{t}} \|(\mathbf{M}_l \delta \mathbf{t})_{rp} - (\boldsymbol{\tau}_{res}^{cmd})_{rp}\|_2^2$$

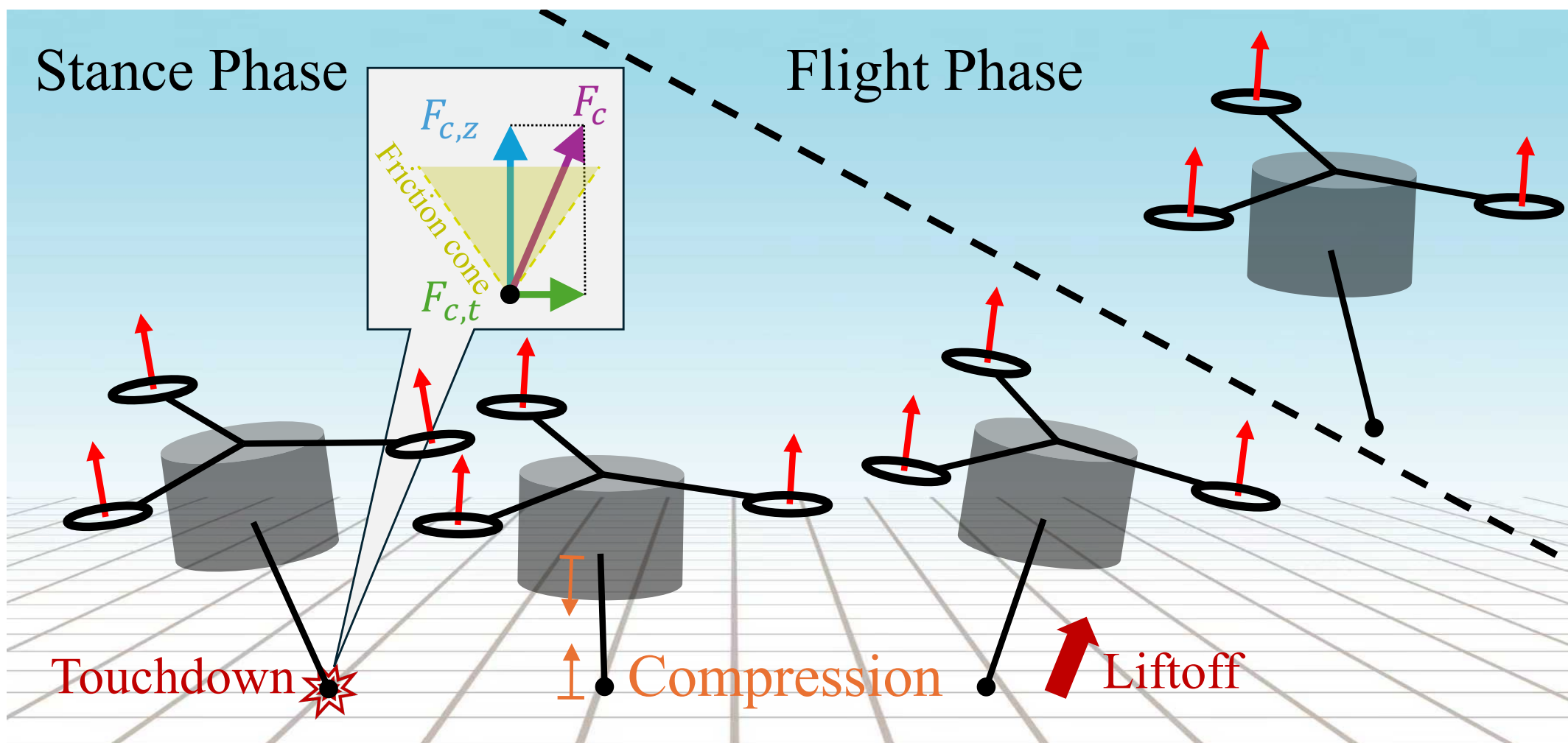
$$\text{s.t. thrust bounds; } \mathbf{t}^{cmd} = \frac{T_{base}}{3} \mathbf{1}_3 + \delta \mathbf{t}^{cmd}, T_{base} = 0.02 \text{ mg.}$$

E. Flight - Raibert foot placement, rotors as primary attitude actuator

$$\mathbf{f}_c = 0; \quad \text{rotors track } (\boldsymbol{\tau}^{des})_{rp}; \quad \text{leg tracks } \mathbf{r}_{td,xy}^{des} = k_v \mathbf{v}_{xy} - k_r \mathbf{v}_{xy}^{des} \text{ toward the next touchdown.}$$

By construction the rotors can only add to - never replace - the leg's contact action.

6. Onehop - event-driven authority handoff

**Touchdown:** $l \leq l_0 - \Delta_{td}$

The leg absorbs impact; the energy law drives the next apex.

Liftoff: $l \geq l_0$

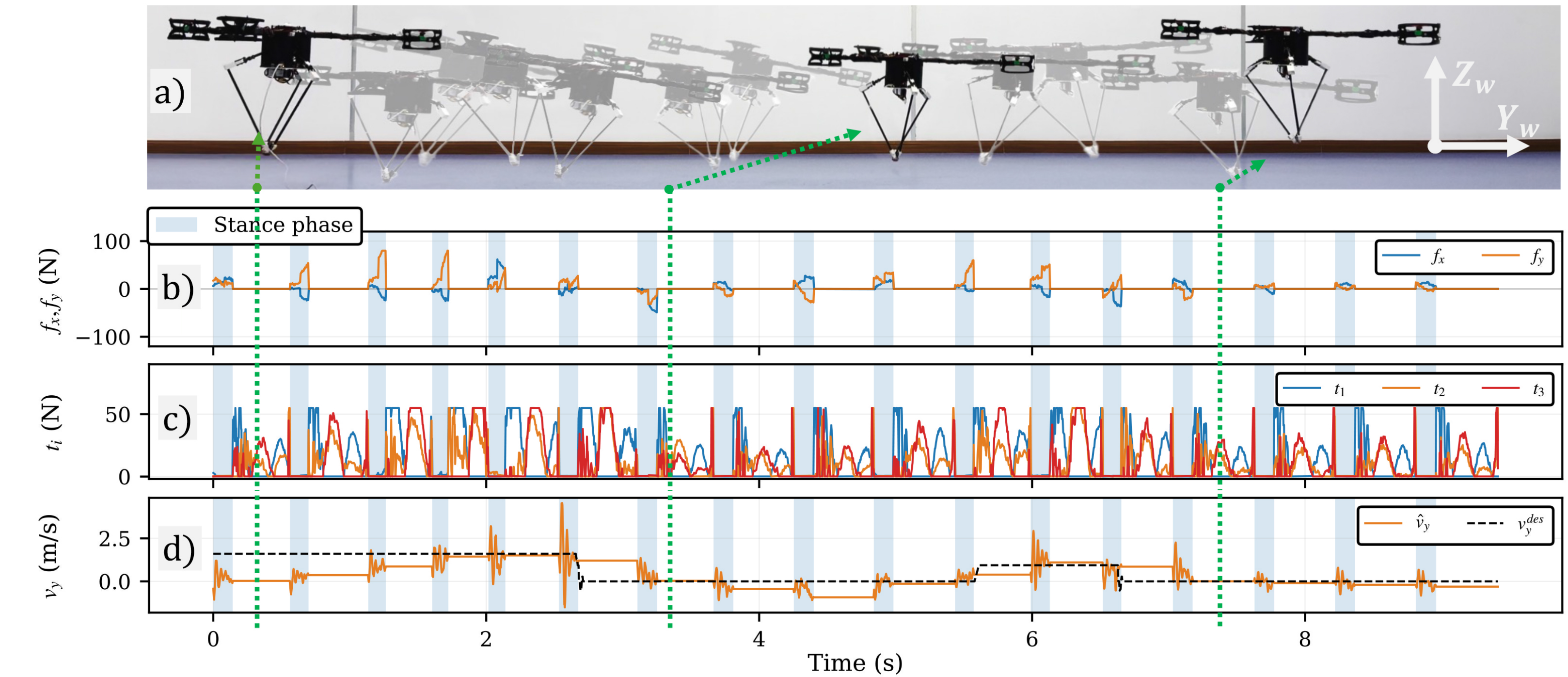
Contact wrench vanishes; rotors become primary for attitude.

Handoff

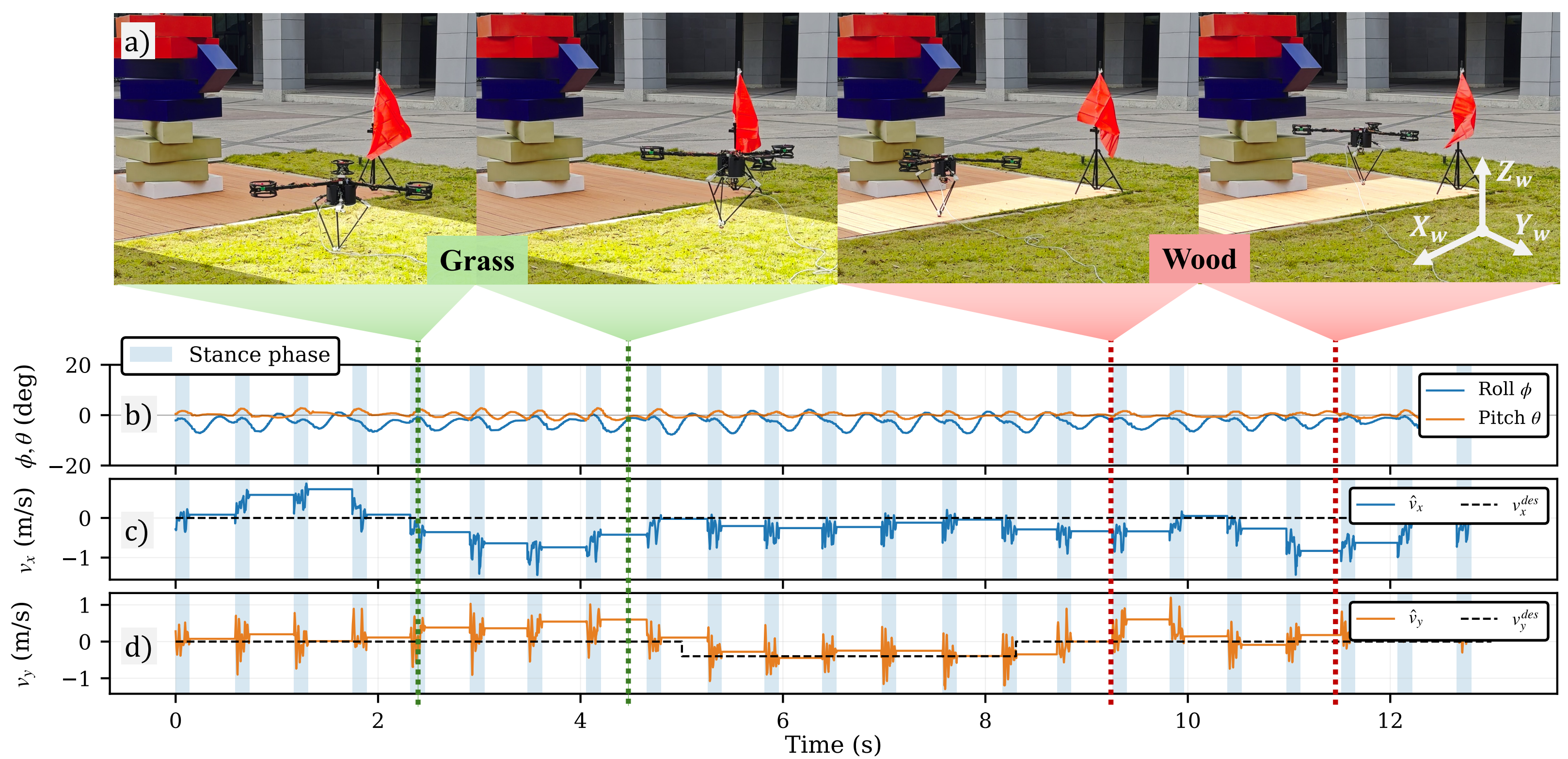
Events switch authority; no hardware or gain change.

Runs in real time: 500 Hz PC control LCM/Ethernet RTT ≈ 0.2 ms Pi: CAN to leg, MSP to flight controller**Leg authority exists only in stance; rotor attitude authority is continuous through the full hop.**

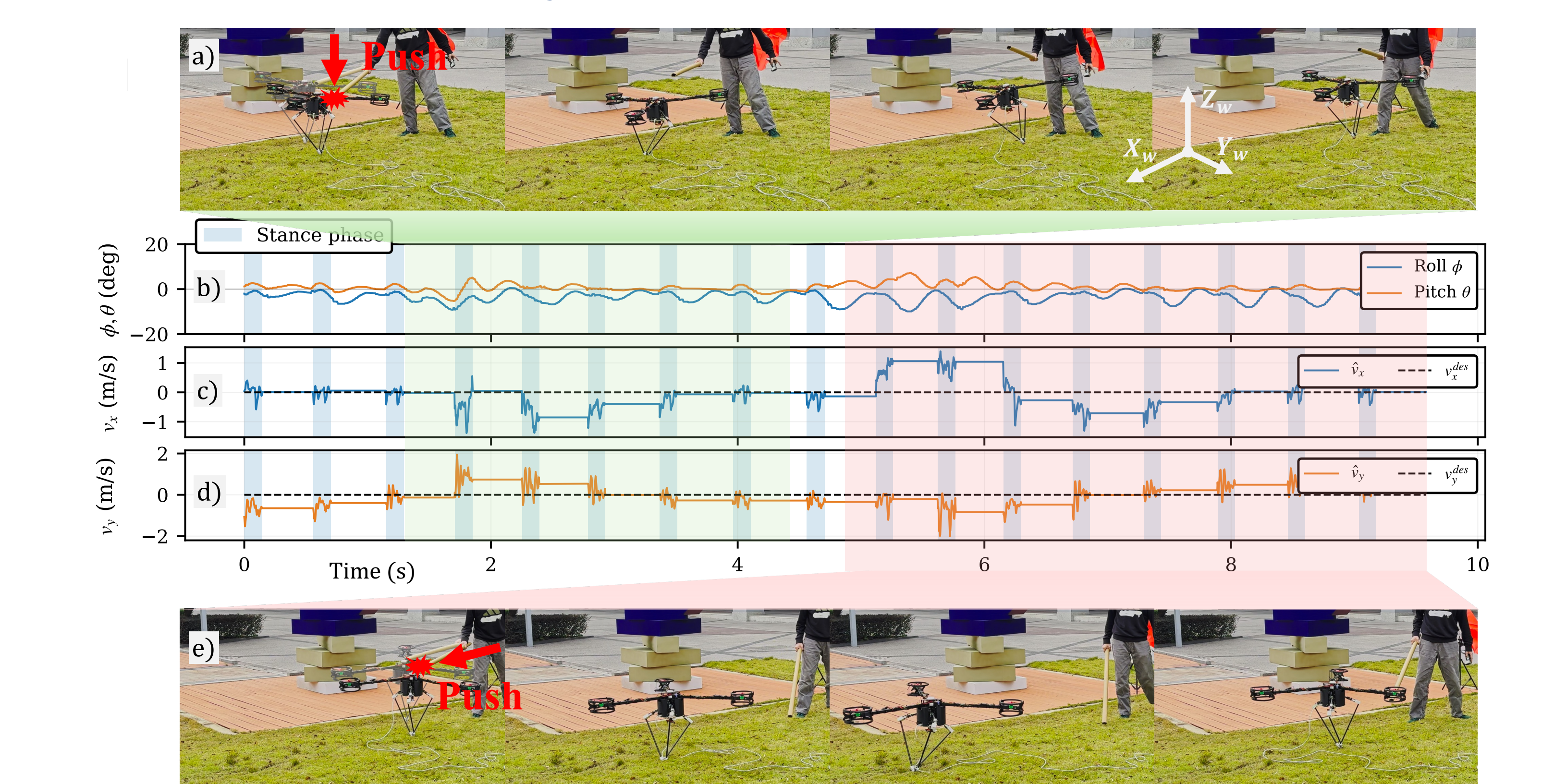
7. Validation - indoor tracking

**17 hops - 1.6 and 0.94 m/s commands - bounded attitude error through every impact.**

8. Validation - grass to wood

**23 hops - same controller - no terrain model - no gain retuning.**

9. Validation - push recovery

**Rotor-arm hit: attitude recovers in 1-2 hops. Main-body hit: lateral velocity recovers in three hops.**